

Lai Jun Jie Clement

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EDUCATION

DIGIPEN INSTITUTE OF TECHNOLOGY (SIT) | SINGAPORE

Sept 2024 – Present

Bachelor of Science in Real-Time Interactive Simulation (expected graduation 2028)

- Core Coursework: Achieved 'A' grades in team-based development modules (Software Engineering Project 3 and 4), Modern C++ Design Patterns, and Design Innovation.

NANYANG POLYTECHNIC | SINGAPORE

Apr 2019 - Feb 2022

Diploma in Cybersecurity & Digital Forensics

- Key Modules: Final Year Project (Grade 'A'), Robotics Process Automation and Applied Cryptography

SKILLS

- Languages: C, C++, C#, GLSL, x86-64 Assembly
- Graphics & Engine: OpenGL, ECS architecture, Deferred Rendering, RapidJSON, ImGui, Mono, FMOD
- Systems: Custom Memory Allocator, Spatial Partitioning
- Math & Algorithms: Vector/matrix Math, Graph algorithms, Kinematics, Pathfinding
- Tools & Workflows: CMake, Git / Git LFS

PROJECTS

DONUT ENGINE | Custom Cross-Platform C++ Game Engine

Core Engine Developer

Sept 2025 – Apr 2026

- Architected a high-performance Entity Component System (ECS) within a custom C++ engine deployed across Windows, macOS and Linux platforms.
- Integrated a Mono C# scripting backend featuring hot-reloading, drastically accelerating gameplay iteration and developer testing workflows.
- Built robust developer tooling utilizing ImGui, including a custom Asset Browser and an intuitive interface for managing Entity Prefabs and Archetypes.
- Engineered and shipped "BEAT EM UP", a 2D action side-scroller rhythm game, serving as the primary technical proof-of-concept for Donut Engine's capabilities.

WORK EXPERIENCE

DIGIPEN INSTITUTE OF TECHNOLOGY (SIT) | SINGAPORE

Lab Teaching Assistant

Jan 2026 - Present

- Facilitated hands-on programming labs and provided real-time debugging support to Year 1 and 2 student teams, actively unblocking technical roadblocks to ensure group projects stayed on track.

BEYONDSOFT | SINGAPORE

Software Test Engineer Intern

May 2021 - Aug 2021

- Conducted integrated software and hardware functionality testing, ensuring accurate calibration of print outputs prior to final deployment.
- Documented bug reports and follow up on software and hardware issues, maintaining an organised and efficient deployment pipeline.

TRANSITLINK | SINGAPORE

Customer Service (Part-time)

Jan 2019 - Feb 2019

- Demonstrated deep user empathy and clear communication by maintaining a proactive, can-do attitude while troubleshooting user issues in fast-paced, high-volume environment.